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Energy Procedia 50 (2014) 581 – 588

Energy Procedia

The International Conference on Technologies and Materials for Renewable Energy, Environment and Sustainability, TMREES14

Feasibility study and electric power flow of grid connected photovoltaic dairy farm in Mitidja (Algeria)

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Abstract

This study presents a feasibility analysis of grid connected photovoltaic system introduction to a dairy farm located in Mitidja plain (north Algeria) to promote farming sustainably. Real load data from ITELV BabaAli dairy farm of 26 dairy cows is used. The farm electricity consumption is mainly for milk production equipments and the cattle housing. The system is optimized by HOMER software using Mitidja global irradiation and the farm load profile. The electricity tariffs and the photovoltaic system components are modelled according to the Algerian market. Viability was determined according to the net present cost, electricity power flow and CO_2 saving potential. According to the optimization criteria, two optimal systems that meet the dairy farm load of 23.6kWh/day have been found. The first optimized system offers a renewable electricity with an optimal cost of 9.7c$/kWh, and the second optimized system has a net annualized surplus electricity of 26.45MWh and a greenhouse gases mitigation of 554 tons over the system lifetime.

Selection and peer-review under responsibility of the Euro-Mediterranean Institute for Sustainable Development (EUMISD)

Keywords: Grid-connected PV system; Dairy farm; Power flow; Sustainable agriculture; HOMER software; Algeria

1. Introduction

With the rapid growth in electricity demand and the depletion of fossil fuel, the Algerian government has realized the importance of renewable energy to prolong fossil fuels reserves lifetime and bring sustainable solutions for combating global climate change, especially greenhouse gas emissions[1].

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doi:10.1016/j.egypro.2014.06.071

Nomenclature

A	Roof area(m^2).	AC	Alternating current.
CoE	Cost of the energy(DA/kWh).	CRF	Capital recovery factor.
DA	Algerian dinars(1$=81DA).	DC	Direct current.
E_{PV}	Annual produced energy(kWh/year).	E_T	Daily load consumption(kWh/day).
F_{GC}	Global correction factor.	G	Global tilted solar radiation(kWh/m^2).
i	Annual real interest rate(%).	IC	Initial set up cost (DA).
$M\&O$	Maintenance and operating costs(DA/year).	N	Number of years.
NPC	Net present cost(DA).	PPC	Purchasing power costs(DA/kWh).
PV	Photovoltaic.	TAC	Total annualized cost(DA/year).
P_C	PV capacity(kWp).	η	PV modules efficiency(%).

Currently, Algeria envisages the substitution of the fossil energies by other alternative sources. National new energetic and environmental policy has ambitious objectives of energy efficiency and renewable energy promotion. The targets of the Algerian policy is to achieve a share of 37% of solar energy and 3% of wind energy in the electricity supply by 2030[2].

Owners of large agricultural buildings with relatively important roofs and often interested by the diversification of their energy resources, farmers have a very particular role to play in the development of the photovoltaic solar installations, dairy farms could produce their own renewable energy through the use of solar photovoltaic. and potentially export photovoltaic green electricity to the power grid during peak electricity demand periods and play a significant role in reducing environmental pollutants as well as lowering farm production costs.

2. Dairy farm characteristics

2.1 Site presentation

Mitidja is a north Algerian plain situated between 36.4° and 36.9° north latitude and 1.7° and 4.1° east longitude with an area of 1450 km^2, it is distributed among four wilayas (Algerian department) namely Algiers, Blida, Tipaza and Boumerdes. Mitidja has a Mediterranean climate and fertile agricultural lands making it one of the most important dairy basin in Algeria.

Our case study is a dairy farm located in the Algerian technical institute of breeding (ITELV) in BabaAli (Algiers), Fig. 1. shows the geographic location of Mitidja. Situated at 36.38° latitude and 3.02° longitude. BabaAli lies on the central part of Mitidja plain and, therefore, displays many of the geographical properties of north Algerian regions.

The farm includes livestock building of 350m^2 that houses 26 dairy cows, a milking parlor for the milk production and a grazing land. The favorable grazing conditions makes the cow's feed based primarily on pasture in spring/summer season and corn silage and hay in autumn/winter, concentrated nutritional supplements are prepared on the farm to meet cows' alimentary needs.

2.2 Farm electricity consumption

Electric equipments information were collected by field investigation and by interviewing farmers about electricity usage patterns, while data about electrical power and consumption were obtained by the monthly energy bills. Total annual electrical energy use for the dairy farm was 8.6MWh, this energy can be classified into four main activities which are milking, milk cooling and storage, farm lighting and cattle feeding as showed in Fig. 2.

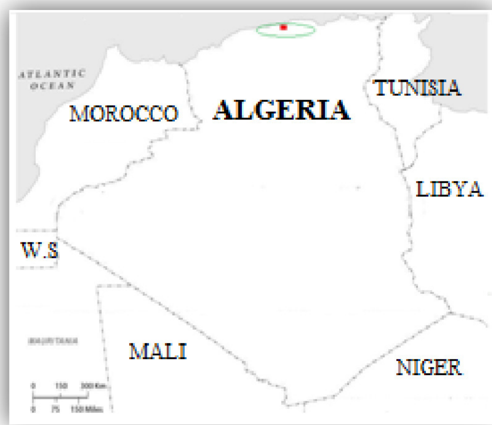


Fig. 1. Geographic location of Mitidja.

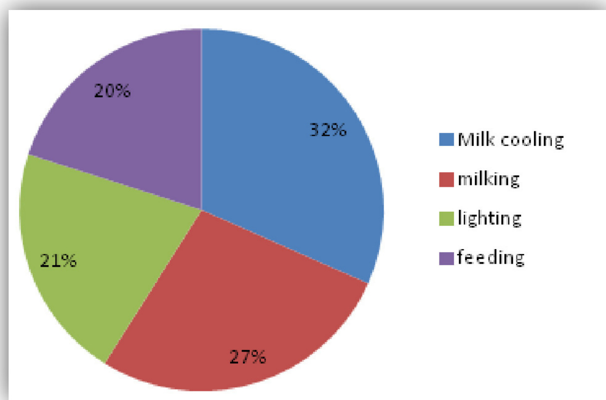


Fig. 2. Electrical energy consumption on the dairy farm.

3. HOMER software

HOMER (Hybrid Optimization Model for Electric Renewable)[3] is a computer optimization model developed by the U.S. National Renewable Energy Laboratory (NREL) to assist the design of micro-power systems and to facilitate the comparison of power generation technologies across a wide range of applications. HOMER models a renewable energy system's behavior, its life-cycle cost and allows to choose optimal design options based on technical, economical and environmental criteria.

3.1 assessment criteria

To judge the feasibility of the grid connected PV system, technical and economical evaluation criteria can be applied. HOMER first assesses the technical feasibility of the system and whether it can meet the load demand, technical feasibility depends mainly on the load energy consumption and the available area to place the photovoltaic modules. Then economical parameters are calculated. The calculation assesses all costs occurring within the project lifetime, including initial set-up costs (*IC*), component replacements within the project lifetime, maintenance and operating costs(*M&O*) and the purchasing power costs (*PPC*) from the grid. The life cycle cost of the system is represented by the net present cost (*NPC*) and calculated according to the following equation[4]

$$NPC = \frac{TAC}{CRF} \tag{1}$$

Where TAC is the total annualized cost (\$/year) (which is the sum of the annualized costs of each system component) and CRF is the capital recovery factor given by

$$CRF = \frac{i(1+i)^N}{(1+i)^N - 1} \tag{2}$$

Where N is the number of years and i is the annual real interest rate (%). The cost of energy produced by the system (CoE) is used also as techno-economical evaluation criterion. CoE represents the unit cost of the energy produced by the system and can be calculated by dividing the total annualized cost by the annual produced energy as follows

$$CoE = \frac{TAC}{E_{PV}} \tag{3}$$

HOMER evaluate a range of equipments and determinate the optimal configuration from the given search space according to the criteria cited above.

4. Data inputs

4.1 Solar irradiance

The solar radiation data for the ITELV farm(36.65° latitude and 3.04° longitude) was obtained from the NASA Surface Meteorology and Solar Energy (SSE) database[5]. The global horizontal solar radiation ranges from 2.1kWh/m^2/day to 6.8kWh/m^2/day and the annual average of the solar radiation is estimated to be 4.51 kWh/m^2/day with a clearness index of 0.55 which make it an ideal location for photovoltaic application.

The 22-year average monthly solar radiation data is illustrated in Fig. 3, which shows the solar radiation data inputs on the left axis and the solar radiation's clearness index on the right axis.

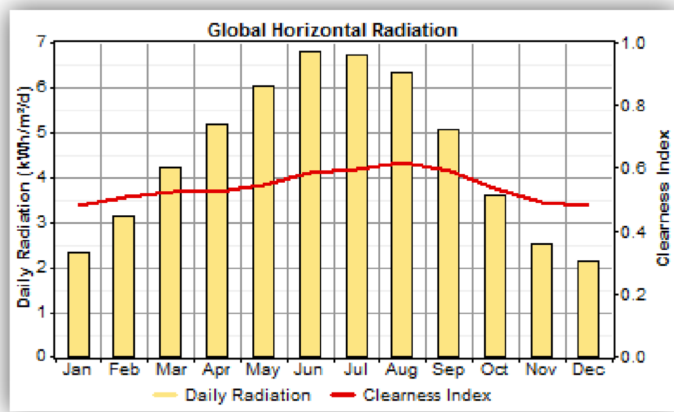


Fig. 3. Global horizontal solar radiation of the farm location[5].

4.2 Load profile

The daily load profile was estimated from the monthly electricity consumption bills. However, the hourly load profile for the whole year cannot be accurately assessed, HOMER allows to introduce random variation to model more realistic load profile. A random variation of 20% and 15% is chosen for the simulation for hourly and daily time step respectively.

Load data showed that the average daily energy consumption is 23.6 kWh and the peak demand is 7.9 kW. Daily profiles of typical spring/summer season and autumn/winter season are illustrated in Fig. 4.

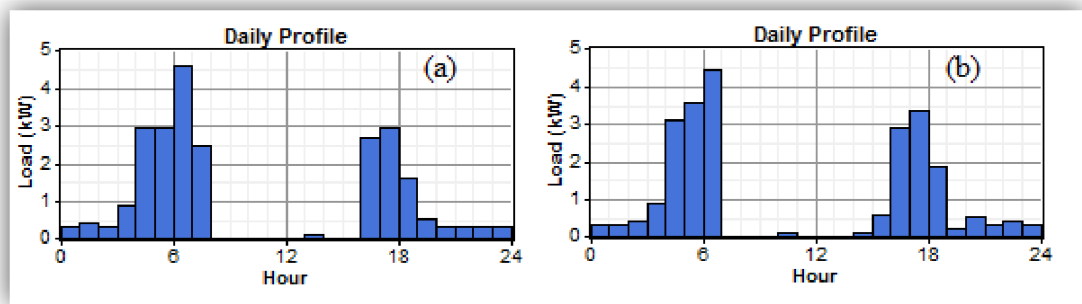


Fig. 4. Load profile of the farm:(a)winter/autumn season, (b) summer/spring season.

4.3 System components

Grid connected PV system consists of a PV array and an inverter, Fig. 5. shows the grid connected photovoltaic system as modeled by HOMER. Generally grid connected systems do not use storage components, the load receives power via the inverter which converts the direct current produced by solar modules into alternating current. The surplus photovoltaic electricity is sold to the grid, and in case of insufficiency of photovoltaic energy, the farm uses the grid to meet the lack of power.

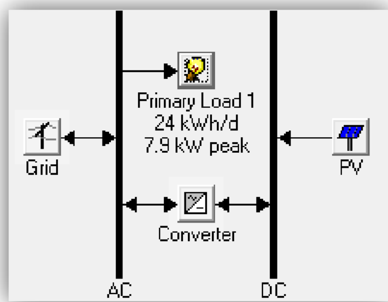


Fig. 5. Grid connected photovoltaic system components.

4.3.1 PV array

The photovoltaic modules used in the feasibility study are the ALPV230 made in Algeria by the ALPVcompany[6], The lifetime of the photovoltaic modules are estimated to 25 years, this period is taken as total project life, a correction factor equal to 0.9 is considered to model 10% of losses due to dust, module aging, loss of cables and all differences between the simulated and the real conditions. Table 1 shows the PV modules technical and financial characteristics.

Table 1. ALPV230 module characteristics.

Technical specifications		*Financial specifications*	
PV module rated power	230Wp	Initial cost	450000DA
Maximum voltage	29.32V	Maintenance cost	2000DA
Maximum current	7.84A	Lifetime	25years
Open circuit voltage	37.17V		
Short circuit current	8.31A		
Efficiency	13.86%		
Dimensions	1663mm x 998mm		

A pre-sizing procedure of the photovoltaic array is performed to limit photovoltaic capacity boundaries and decrease the search space. The lower boundary which guarantees that the photovoltaic electricity meets the load consumption is determined according to the following equation

$$P_C = \frac{E_T}{G.F_{GC}} \tag{4}$$

With Pc is the photovoltaic capacity, E_T is the daily load electricity consumption, G is the global tilted solar radiation and F_{GC} is the global correction factor (0.6-0.9).
The upper boundary depends on the available roof area and can be determined according to the equation

$$P_C = \eta.A \tag{5}$$

With η the photovoltaic modules efficiency and A represents the roof area. The roof area of the studied farm geometry is 7.24mx25m with a tilt of 15°. The maximum number of PV modules that can be mounted on the roof is 100(4x25).Using this boundaries pre-sizing the search space of the PV array will comport all power capacity between 5.75kWpeak and 23kWp with a step of 230Wp.

4.3.2 Converter

A grid connected inverter is required for PV system to maintain the flow of energy between DC photovoltaic generation and AC load and power grid. In this system, three single-phase ABB Power-one^R grid connected inverters are included in the search space, it is assumed that the inverters have an efficiency of 95% and 20 years lifetime. The inverters must be replaced once during the project period and the replacement cost is supposed 50% of the initial cost . The technical and economic data of the inverters are shown in Table 2. The maximal input current of the inverters(20A) is bigger than twice the maximum output current of the ALPV230 photovoltaic module(7.84A) then the photovoltaic space search step will be 0.46Wp.

Table 2. Inverters characteristics.

	PVI 4.2	PVI 5.0	PVI 6.0
Input DC power (W)	4375	4800	6200
Input maximal DC voltage (V)	600	600	600
Input maximal current (A)	20	22	22
Operating DC voltage (V)	84-580	84-580	84-580
Output AC voltage (V)	230	230	230
Lifetime (years)	20	20	20
Initial cost (DA)	203000	219400	244000
Replacement cost (DA)	101500	109700	122000
Maintenance cost (DA/year)	0	0	0

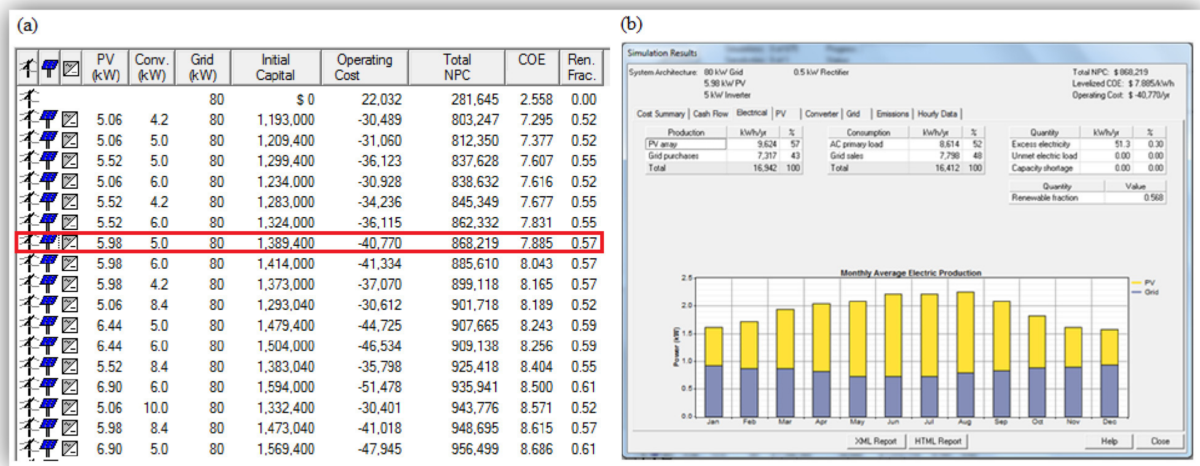
4.3.3 Power grid

Algerian power grid is managed by the National company Sonalgaz, Electricity tariffs are regulated by the Regulation Commission for Electricity and Gas CREG, the most common tariff for agricultural farm is the rate E43 which consist of day rate tariff of 357cDA/kWh (6:00am to10:30pm) and night rate tariff of 85cDA/kWh (10:30 pm 6:00am).

According to the 5 February 2002 law and the 25 march 2004 decree[7], the feed-in tariff of grid connected photovoltaic electricity in Algeria is four times the purchasing price which means 14,28DA/kWh.

5. Results and discussion

HOMER simulates all the search space system configurations and provides as output the feasible systems in an increasing order according to their *NPC*. The optimization results for the dairy farm grid connected photovoltaic system are illustrated in Fig. 6(a). It is clear that the grid connected system with 5.06kWc PV capacity offers the lowest *NPC*, but this configuration does not meet the total annual farm consumption and represents an electrical energy deficit. The grid connected system with 26 photovoltaic modules (5.98kWp) and PVI5.0 inverter is the optimal power system for the dairy farm, it satisfies the farm load and offers 5.6% of surplus electricity annually. The suggested optimal system was found to have an initial cost of 1.39MDA(17,150$), an annual operating cost of -40770DA/year (503$/year), a total *NPC* of 868,219DA(10718$) and a levelized cost of energy of 7.88DA/kWh (9.7c$/kWh), the payback time is estimated to 23 years at Algerian electricity prices. The monthly average electric production from the optimized system is shown in Fig. 6(b) and the detailed power flow between the system and the utility grid is summarized in Table. 3. The total annual photovoltaic electricity production from the optimized system is 9624kWh with 7798kWh (81%) sold to the grid, 1296kWh (13.5%) consumed by the farm and 530kWh (5.5%) lost due to inverter efficiency and wires modeled losses.


(a)

PV (kW)	Conv. (kW)	Grid (kW)	Initial Capital	Operating Cost	Total NPC	COE	Ren. Frac.
		80	$ 0	22,032	281,645	2.558	0.00
5.06	4.2	80	1,193,000	-30,489	803,247	7.295	0.52
5.06	5.0	80	1,209,400	-31,060	812,350	7.377	0.52
5.52	5.0	80	1,299,400	-36,123	837,628	7.607	0.55
5.06	6.0	80	1,234,000	-30,928	838,632	7.616	0.52
5.52	4.2	80	1,283,000	-34,236	845,349	7.677	0.55
5.52	6.0	80	1,324,000	-36,115	862,332	7.831	0.55
5.98	5.0	80	1,389,400	-40,770	868,219	7.885	0.57
5.98	6.0	80	1,414,000	-41,334	885,610	8.043	0.57
5.98	4.2	80	1,373,000	-37,070	899,118	8.165	0.57
5.06	8.4	80	1,293,040	-30,612	901,718	8.189	0.52
6.44	5.0	80	1,479,400	-44,725	907,665	8.243	0.59
6.44	6.0	80	1,504,000	-46,534	909,138	8.256	0.59
5.52	8.4	80	1,383,040	-35,798	925,418	8.404	0.55
6.90	6.0	80	1,594,000	-51,478	935,941	8.500	0.61
5.06	10.0	80	1,332,400	-30,401	943,776	8.571	0.52
5.98	8.4	80	1,473,040	-41,018	948,695	8.615	0.57
6.90	5.0	80	1,569,400	-47,945	956,499	8.686	0.61

Fig. 6.(a)Optimization Overall results;(b)Electrical production of the optimized system.

Table 3. Detailed electric power flow of the optimized system.

Month	Consumption (kWh)	Produced energy (kWh)	Purchased energy (kWh)	Sold energy (kWh)	Net purchases (kWh)
January	735	515	683	437	246
February	648	569	583	475	108
March	744	793	644	650	-6
April	718	876	591	704	-114
May	703	1006	547	794	-247
June	706	1071	530	835	-305
July	721	1110	543	868	-324
August	765	1087	594	848	-254
September	723	904	603	730	-127
October	729	702	656	594	62
November	787	515	644	446	198
December	734	474	699	416	283
Annual	8,614	9,624	7,317	7,798	-480

The optimized system resulted in a negative total GHG emission, facilitated through sale of surplus electricity to the grid. Overall GHG savings in comparison to grid-only is approximately 5.75 tons CO2 yearly.

Without taking account of the economic performance of the system, the grid connected configuration that provides the maximum energy to the grid and contributes the most to the GHG mitigation is the system with 100 PV modules (23kWp) and four PVI5.0 inverters, the renewable fraction *RF* of this system is 86% and his total annual photovoltaic electricity production is 37MWh with 32,525kWh (88%) sold to the grid and annual net surplus electricity of 26.45MWh .The payback time is reduced to 18.6 years and the overall GHG saving is 22.16 tons CO2 yearly which means 554 tons of CO2 in the project lifetime. The variation of the electrical net energy flow and CO2 mitigation according to the PV system capacity(expressed by the PV modules number) is illustrated in Fig. 7.

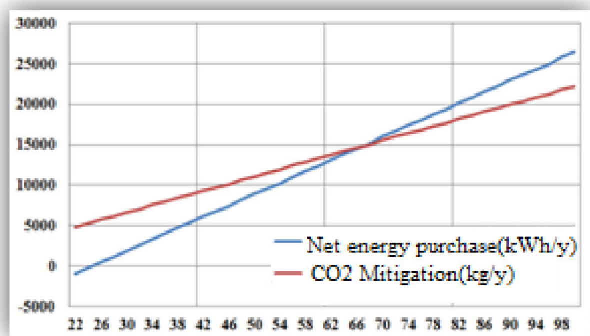


Fig. 7.Net surplus electricity and GHG mitigation potential variation with the PV modules number.

6. Conclusion

A feasibility study of photovoltaic grid connected system introduction to the Mitidja's dairy farm has been presented. The analysis used real load data from ITELV BabaAli dairy farm(26 dairy cows) located in north Algeria. HOMER software was used for feasibility assessment, and viability was determined according to net present cost (*NPC*), power flow obtained from the net purchased electricity, payback time and the GHG mitigation potential of the system. The system components used in the simulation are available in the local market and the economical analysis is performed using Algerian market prices. It is found that the optimal PV system has a levelized cost of energy (*CoE*) less than 0.1$/kWh, 23 years payback time and 5.75tons of CO2 savings potential. These values could be improved by increasing the PV system capacity up to the maximum feasible limit which is 23kWp, but it will also increase the total *NPC* and the cost of energy.

Acknowledgements

This work is performed in collaboration with the Algerian Technical Breeding Institute according to the convention n° 630/DG/2012. The authors would like to think Dr.Nadjemi H, Dr.Sakhri L and Mr.ATIF M E. for their helpful assistance in the dairy farm data collection.

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